Chapter 2 - The Real Number System

Note 4

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2.4 : MATHEMATICAL INDUCTION

In this section, we shall revisit the method of **Mathematical Induction** and use it to prove the **Binomial Formula**, a result that shows how to expand powers of a binomial expression. We begin by obtaining another consequence of the Completeness Axiom, the ***Well-Ordering Principle***, which is a statement about the existence of least elements of subsets of $\mathbb{Z}^+$.

Theorem 2.11 : Well-Ordering Principle

If E is a non-empty subset of $\mathbb{Z}^+$, then E has a least element (i.e., E has a finite infimum and $\inf E \in E$).

Proof.

- Suppose that $E \subseteq \mathbb{Z}^+$ is non-empty.
- Then $-E$ is bounded above, by -1, so by the **Completeness Axiom** $\sup(-E)$ exists, and by **Theorem 2.6**, $\sup(-E) \in -E$.
- Hence by **Theorem 2.9**, $\inf E = -\sup(-E)$ exists, and $\inf E \in -(-E) = E$. □

Our first application of the **Well-Ordering Principle** is called the Principle of Mathematical Induction.

Theorem 2.12 : Suppose for each $n \in \mathbb{Z}^+$ that $A(n)$ is a proposition which satisfies the following two properties:

(i). $A(1)$ is true.

(ii). For every $n \in \mathbb{Z}^+$ for which $A(n)$ is true, $A(n+1)$ is also true.

Then $A(n)$ is true for all $n \in \mathbb{Z}^+$.

The following examples illustrate the fact that the Principle of Mathematical Induction can be used to prove a variety of statements involving integers.

Example Prove that $\sum_{k=1}^{n}(3k-1)(3k+2) = 3n^3+6n^2+n \quad$ for $n \in \mathbb{Z}^+$.

Proof.

- Let $A(n)$ represent the statement

$$\sum_{k=1}^{n}(3k-1)(3k+2) = 3n^3+6n^2+n \ .$$

- For $n = 1$, the left side of this equation is $2\cdot 5 = 10$ and the right side is $3+6+1 = 10$. Therefore $A(1)$ is true.

- Suppose that $A(n)$ is true for some $n \geq 1$. Then we have

$$\begin{aligned}\sum_{k=1}^{n+1}(3k-1)(3k+2) &= (3n+2)(3n+5) + \sum_{k=1}^{n}(3k-1)(3k+2) \\ &= (3n+2)(3n+5) + 3n^3+6n^2+n \\ &= 3n^3+15n^2+22n+10 \ .\end{aligned}$$

- On the other hand, a direct calculation reveals that

$$3(n+1)^3+6(n+1)^2+(n+1) = 3n^3+15n^2+22n+10$$

- Therefore, $A(n+1)$ is true when $A(n)$ is.

- We conclude by induction that $A(n)$ holds for all $n \in \mathbb{Z}^+$. □

Two formulas you have already encountered in mathematics are the perfect square and cube formulas:

$$(a+b)^2 = a^2+2ab+b^2 \quad \text{and} \quad (a+b)^3 = a^3+3a^2b+3ab^2+b^3 \ .$$

Our next application of the **Principle of Mathematical Induction** generalizes these formulas from $n = 2$ and 3 to arbitrary $n \in \mathbb{Z}^+$.

The **Pascal's triangle** is a triangular array of integers whose rows begin and end with 1s with the property that an interior entry on any row is obtained by adding the two numbers in the preceding row immediately above that entry. Thus the first few rows of Pascal's triangle are as below:

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1
\end{array}
$$

Notice that the third and fourth rows are precisely the coefficients that appeared in the perfect square and cube formulas above.

We can write down a formula for each entry in each row of the Pascal triangle. The first (and only) entry in the first row is $\binom{0}{0} := 1$.

Using the notation $0! := 1$ and $n! := 1 \cdot 2 \cdots (n-1) \cdot n$ for $n \in \mathbb{Z}^+$, define the binomial coefficient n choose k by

$$\binom{n}{k} := \frac{n!}{(n-k)!\ k!} \quad \text{for} \quad 0 \leq k \leq n \quad \text{and} \quad n = 0, 1, \ldots \ .$$

Since $\binom{n}{0} = \binom{n}{n} = 1$ for all $n \in \mathbb{Z}^+$, the following result shows that the binomial coefficient n over k does produce the $(k+1)^{\text{st}}$ entry in the $(n+1)^{\text{st}}$ row of Pascal's triangle.

Lemma : If $n, k \in \mathbb{Z}^+$ and $1 \leq k \leq n$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof. We shall prove that the left hand side equals to the right hand side which directly follows from the definition as follows:

$$\begin{aligned}
\binom{n}{k-1} + \binom{n}{k} &= \frac{n!\ k}{(n-k+1)!\ k!} + \frac{n!\ (n-k+1)}{(n-k+1)!\ k!} \\
&= \frac{n!\ (n+1)}{(n-k+1)!\ k!} \\
&= \binom{n+1}{k} .
\end{aligned}$$

□

Binomial coefficients can be used to expand the nth power of a sum of two terms.

Theorem 2.13 : Binomial Formula

If $a, b \in \mathbb{R}, n \in \mathbb{Z}^+$ and 0^0 is interpreted to be 1, then

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

Proof of the above theorem is left as an Exercise (See **Exercise 1**).

We close this section with two well-known results that further demonstrate the power of the Completeness Axiom and its consequences.

Result 1 : If $x > 1$ and $x \notin \mathbb{Z}^+$, then there is an $n \in \mathbb{Z}^+$ such that $n < x < n+1$.

Proof of Result 1.

- By the **Archimedean Principle**, the set $E = \{m \in \mathbb{Z}^+ \ : \ x < m\}$ is non-empty.
- Hence by the **Well-Ordering Principle**, E has a least element, say m_0.
- Set $n = m_0 - 1$. Since $m_0 \in E$, $\quad n+1 = m_0 > x$.
- Since m_0 is least, $n = m_0 - 1 \leq x$. Also, since $x \notin \mathbb{Z}^+$, we also have $n \neq x$.
- Therefore, $n < x < n+1$. □

Using this last result, we can prove that the set of irrationals is non-empty.

Result 2 : If $n \in \mathbb{Z}^+$ is not a perfect square (i.e., if there is no $m \in \mathbb{Z}^+$ such that $n = m^2$), then $\sqrt{n}$ is irrational.

Proof of Result 2.

- Suppose to the contrary that $n \in \mathbb{Z}^+$ is not a perfect square but $\sqrt{n} \in \mathbb{Q}$; that is $\sqrt{n} = \dfrac{p}{q}$ for some $p, q \in \mathbb{Z}^+$.

- Choose by **Result 1** an integer $m_0 \in \mathbb{Z}^+$ such that

$$m_0 < \sqrt{n} < m_0 + 1 \ . \tag{2.4.1}$$

- Consider the set $E := \{k \in \mathbb{Z}^+ \ : \ k\sqrt{N} \in \mathbb{Z}\}$.

- Since $q\sqrt{n} = p$, we know that E is non-empty.

- Thus by the **Well-Ordering Principle**, E has a least element, say n_0.

- Set $x = n_0(\sqrt{n} - m_0)$. By **(2.4.1)**, $0 < \sqrt{n} - m_0 < 1$.

- Multiplying this inequality by n_0, we find that

$$0 < x < n_0 \ . \tag{2.4.2}$$

- Since n_0 is the least element of E, it follows from **(2.4.2)** that $x \notin E$.

- On the other hand, since $n_0 \in E$, it follows that

$$x\sqrt{n} = n_0(\sqrt{n} - m_0)\sqrt{n} = n_0 n - m_0 n_0 \sqrt{n} \ \in \mathbb{Z} \ .$$

- Moreover, since $x > 0$ and $x = n_0\sqrt{n} - n_0 m_0$ is the difference of two integers, $x \in \mathbb{Z}^+$.

- Thus $x \in E$, which is a contradiction.

□

EXERCISES

(1). Prove Theorem 2.13. (**Hint:** Use Induction on n and the Lemma in Page 3)

(2). Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples to the false ones.

(a). If $a \geq 0$ and $b \neq 0$, then $(a+b)^n \geq b^n$ for all $n \in \mathbb{Z}^+$.

(b). If $a < 0 < b$, then $(a+b)^n \leq b^n$ for all $n \in \mathbb{Z}^+$.

(c). If $n \in \mathbb{Z}^+$ is even and both a and b are negative, then $(a+b)^n > a^n + na^{n-1}b$.

(d). If $a \neq 0$, then

$$\frac{1}{2^n} = \sum_{k=0}^{n} \binom{n}{k} \frac{(a-2)^{n-k}}{a^n 2^{n-k}}$$

for all $n \in \mathbb{Z}^+$.

(3). (a) Prove that if $x_1 > 2$ and $x_{n+1} = 1 + \sqrt{x_n - 1}$ for $n \in \mathbb{Z}^+$, then $2 < x_{n+1} < x_n$ holds for all $n \in \mathbb{Z}^+$.

(b) Prove that if $2 < x_1 < 3$, and $x_{n+1} = 2 + \sqrt{x_n - 2}$ for $n \in \mathbb{Z}^+$, then $0 < x_n < x_{n+1}$ holds for all $n \in \mathbb{Z}^+$.

(4). Prove that $2^n \leq n! + 2$ for $n \in \mathbb{Z}^+$.

(5). Prove that the following formulas hold for all $n \in \mathbb{Z}^+$.

(a) $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

(b) $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

(c) $\displaystyle\sum_{k=1}^{n} \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, \quad a \neq 0$

(d) $\displaystyle\sum_{k=1}^{n} (2k-1)^2 = \frac{n(4n^2-1)}{3}$